

Broken Nalgene Secondary Container: Lessons Learned

Purpose

This Lessons Learned focuses on an incident involving a spill of multiple chemicals and how proper risk assessment can help turn what may seem like an emergency into a manageable situation. It also highlights the importance of paying close attention to the condition of your lab equipment.

What Happened?

A researcher was moving a plastic Nalgene secondary container that had roughly 20 containers of hazardous materials in it from the benchtop to another location. The researcher was wearing safety glasses, a flame-resistant lab coat, nitrile gloves, and lab-appropriate attire. They lifted the secondary container by the handles with both hands, but after raising it only a few inches, the handle broke. The container fell to the benchtop, where it broke further. This led to some hazardous materials containers falling to the floor and breaking open. Prior to lifting the container, the researcher had seen no signs to suggest the Nalgene was fragile; there were no cracks, discolorations, or other indications of structural weakness.

The researcher considered calling EH&S immediately but decided to take some time to assess the situation. First, they activated the emergency exhaust on the nearby fume hoods to increase ventilation in the area near the spill. Then, they examined the spilled materials. Based on their familiarity with the contents of the secondary container, they identified a 100 gram container of 4-hydroxybenzaldehyde and a roughly 10 mL container of ethyl orthoformate. They were also able to identify that a round bottom flask that had broken had contained roughly 10 mL of Hünig's base due to the distinctive odor of this reagent. The last container's label was not readily visible due to how it had broken, so the researcher carefully turned over the piece of glass with the label and determined that it had been a 1 L container of methanol. The researcher knew that none of these materials would react with each other and that none posed an immediate health hazard, so they decided to clean up the materials themselves. They then notified their PI and submitted an incident report.

**Call EH&S immediately at
(650) 725-9999 if you believe a spill
is health-threatening.**

Why Did This Happen?

The root cause was a structural weakness in the plastic Nalgene secondary container. In general, plastic materials become weaker with time. While such containers may show signs of their degradation such as cracks or discoloration, in some instances there is no obvious sign prior to breakage.

What Went Right?

- The researcher immediately remembered that calling EH&S was an option. While they ultimately decided that assistance was not necessary, it is important to remember that you can call the EH&S incident hotline at (650) 725-9999 if you believe a spill is health-threatening.
- The researcher remembered that their fume hood had an emergency exhaust feature, which allowed them to increase the local ventilation in the area of the spill.

- The researcher conducted a good risk assessment. They carefully ascertained the identity of the materials, evaluated their reactivity and toxicity, and determined that cleanup was a relatively low risk activity.
- The researcher was familiar with the properties of the materials involved, which saved them time and effort in the assessment. The researcher was also familiar and comfortable with the proper procedure for how to clean up spills.
- The materials stored in the secondary container were compatible with each other, so no unwanted chemical reactions occurred.
- The researcher was wearing appropriate PPE. If there had been more liquid materials, they might have splashed, but the researcher's attire would have protected them from direct skin contact.

What Lessons Are There?

- Inspect your secondary containers for defects regularly. Additionally, while this secondary container was not overloaded, avoid storing multiple heavy objects in a single secondary container if possible.
- Always handle plastic containers carefully and while wearing appropriate PPE.
- Be aware of the materials in use around you and, whenever possible, inform yourself of their hazards. Even if you do not work with a given material personally, knowing its hazards will be vital if it spills.
- If you do not feel safe cleaning up a spill, or you feel that it is health-threatening, call the EH&S 24-hour incident hotline at 650 725 9999.
- Do not store materials with relatively higher hazards (e.g., pyrophoric, toxic by inhalation) out on the benchtop, and instead store them in cabinets. Additionally, store materials compatibly so that there are no unintended reactions in the event of a leak or spill.
- If you must manipulate broken glass, use tongs or other tools. Do not handle broken glass with your hands.
- Clean up broken glass using a broom and dustpan. If the glass is uncontaminated, place it in a broken glass container. If it is contaminated, place it in a hard-sided red sharps container.
 - If the contamination was chemical in nature (as opposed to biological), be sure to put a waste tag reflecting the identity of the contaminant(s) on the sharps container.

If you have any questions, please feel free to contact EH&S by calling (650) 723-0448 or submitting a question on our website.